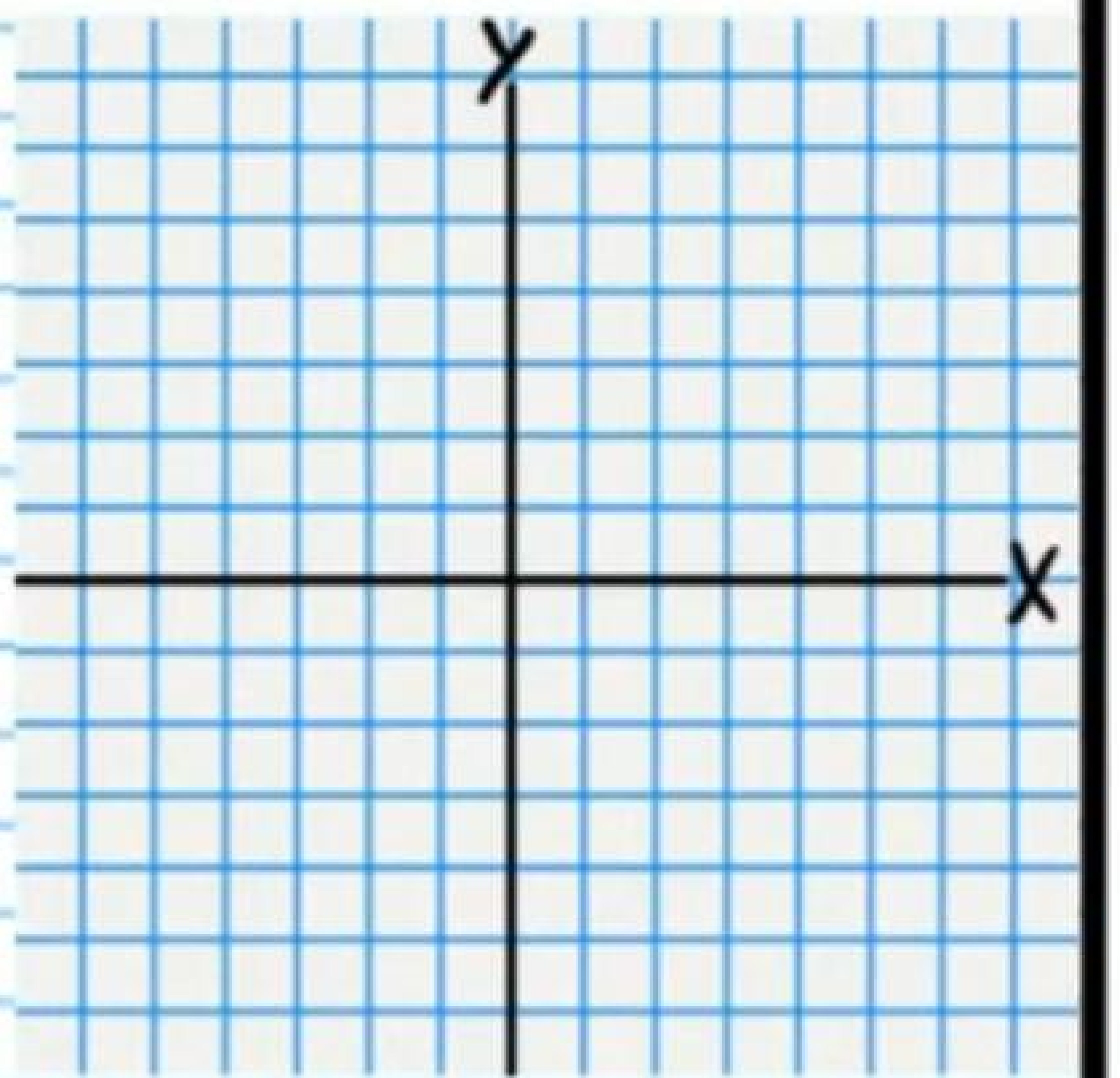
Name
Date
Course

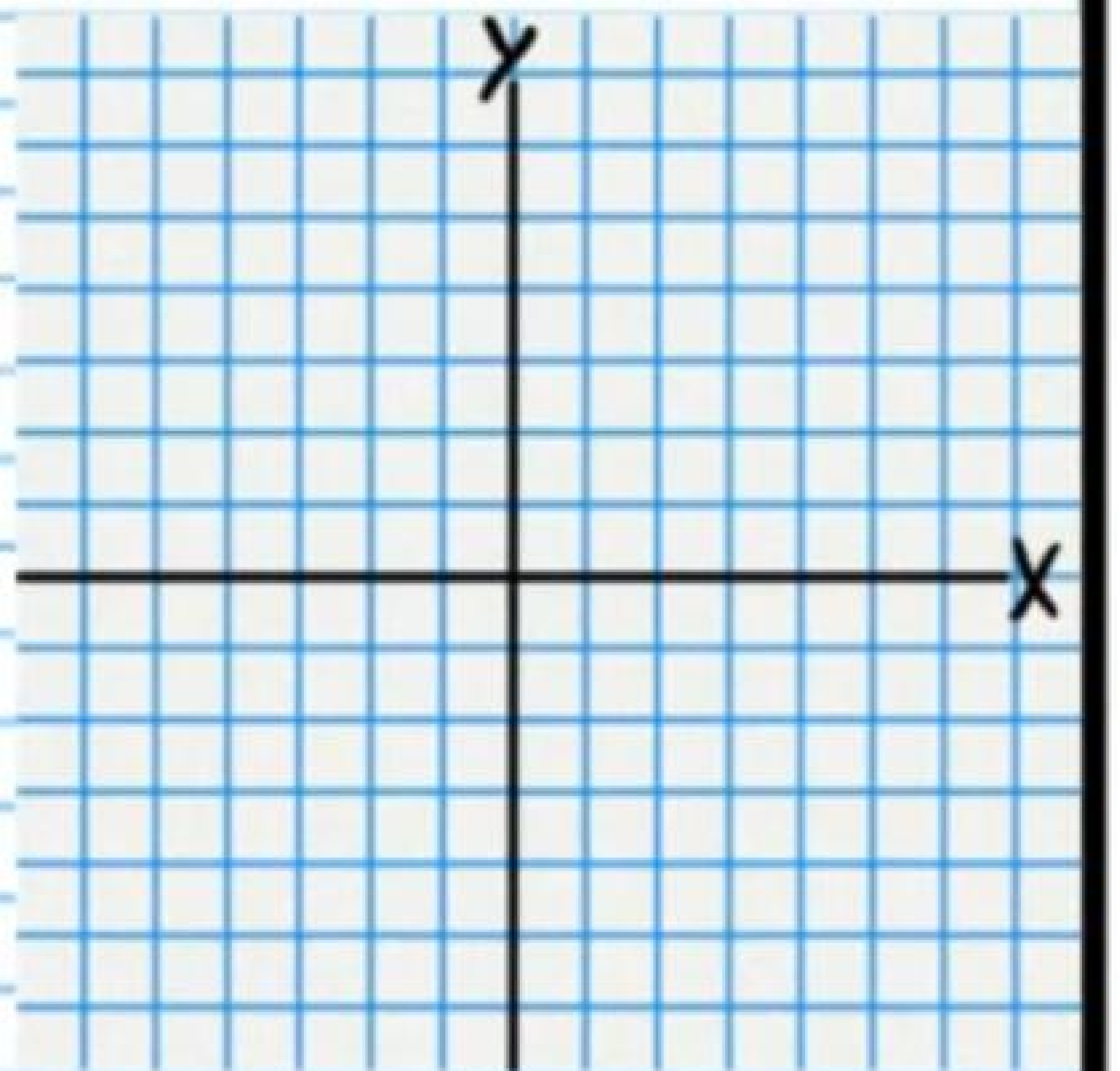
The Conics (Part I) Homework

Graph the following equations. Write the coordinates of the center and the length of the radius.

① $(x+3)^2+(y+3)^2=9$

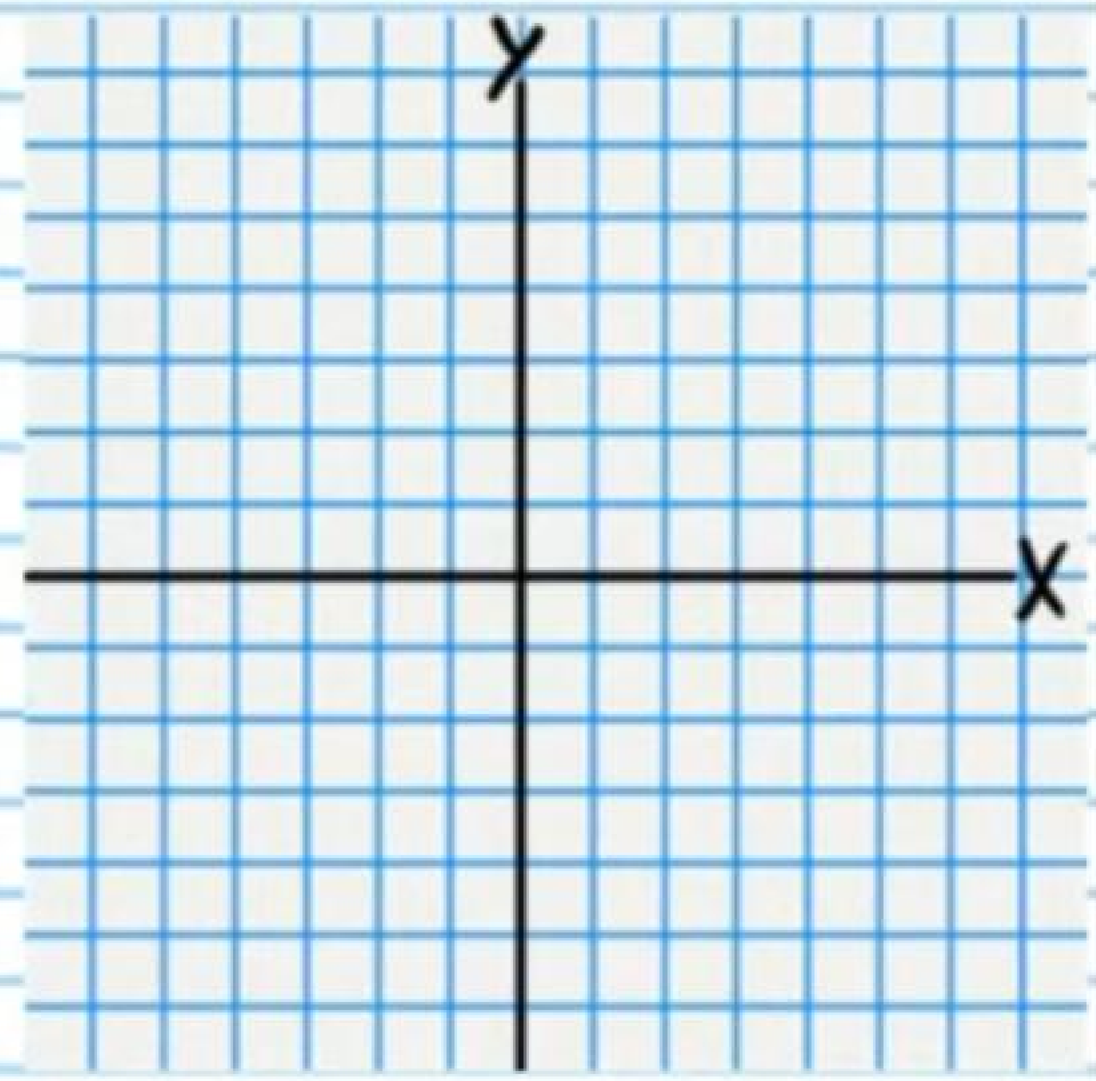


② $4=(x-4)^2+(y+1)^2$

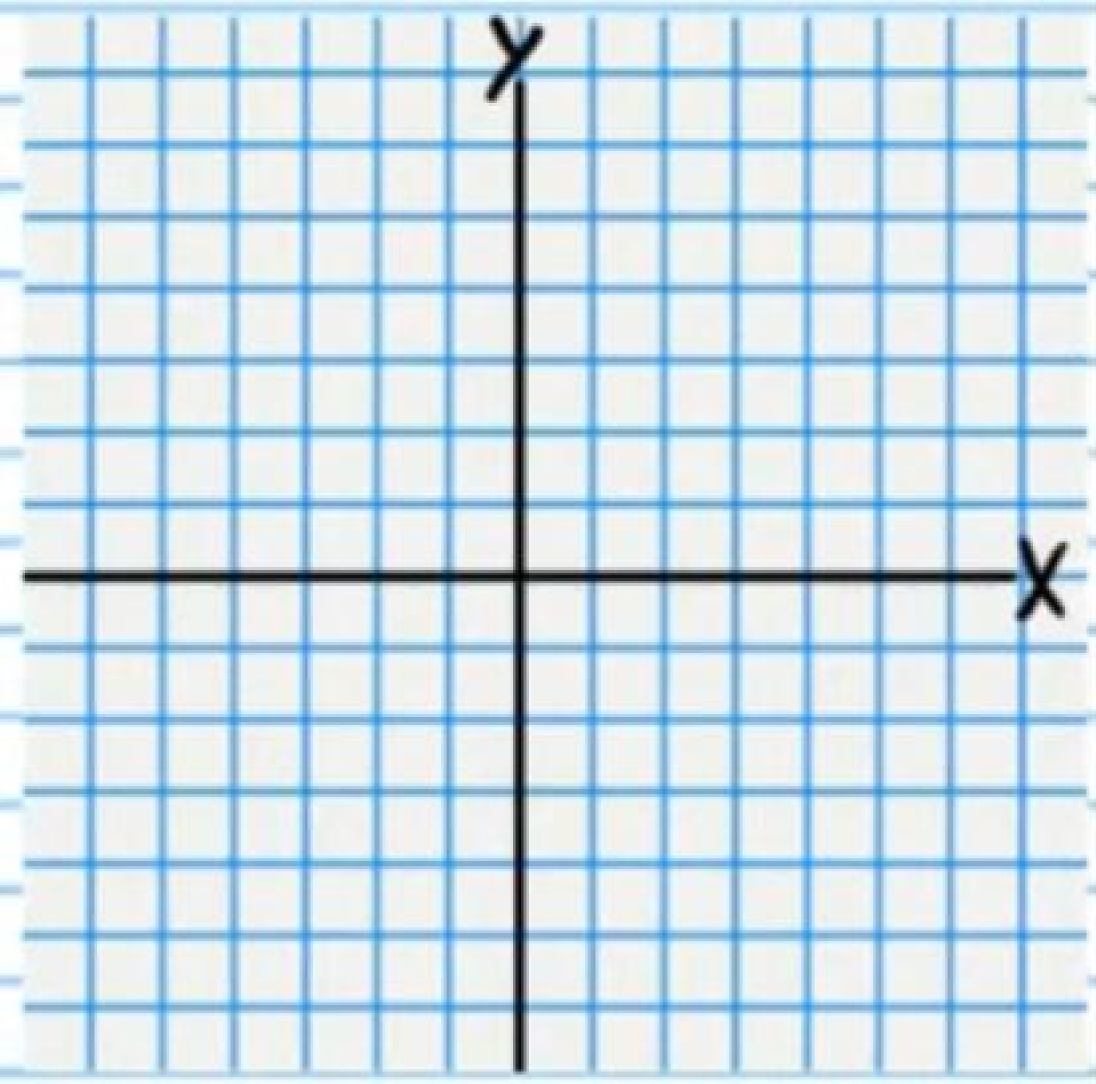


Graph the following equations. Write the coordinates of the focus and the equation of the directrix.

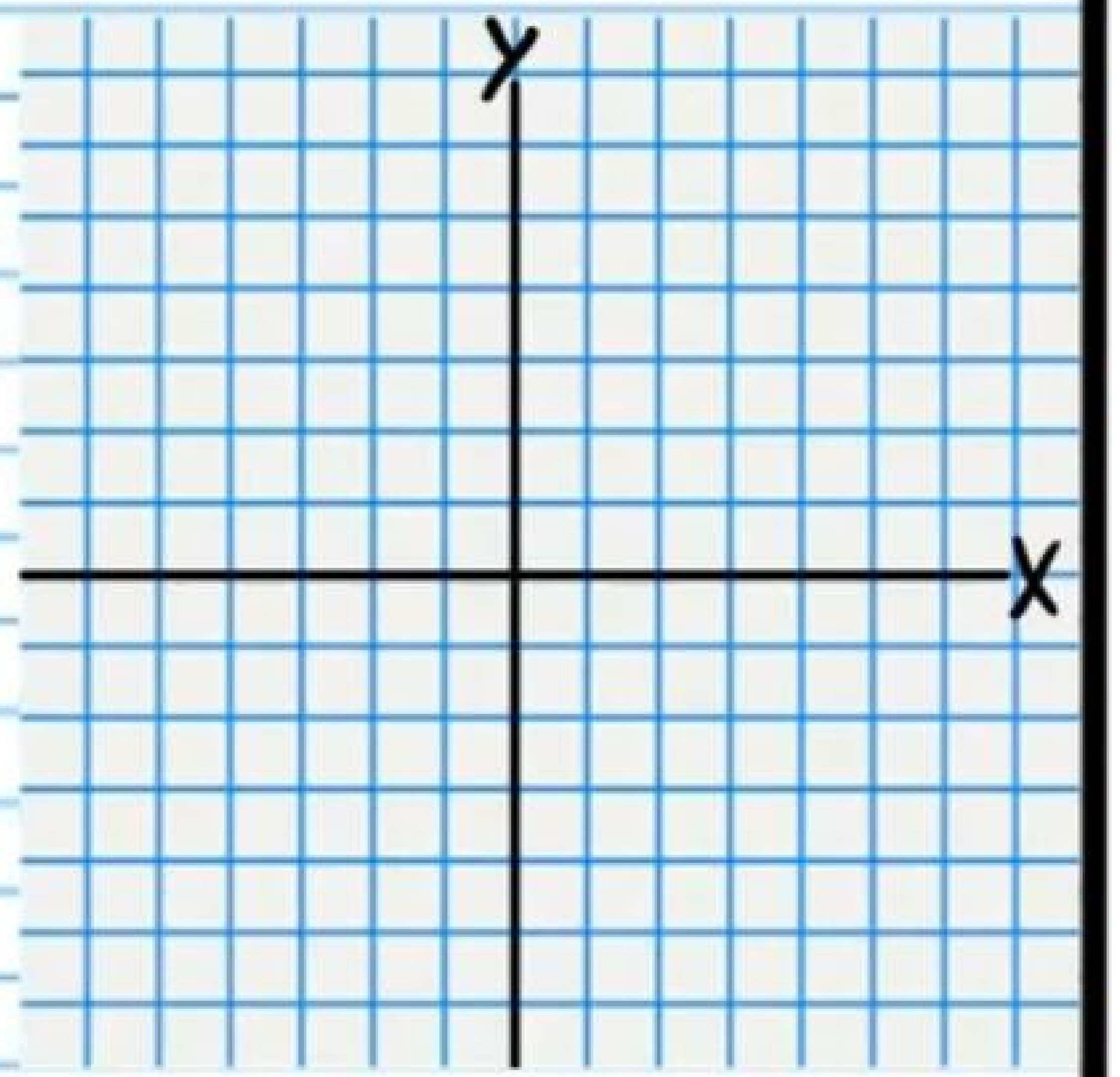
③ $\frac{1}{2}(x+3)^2=y+2$



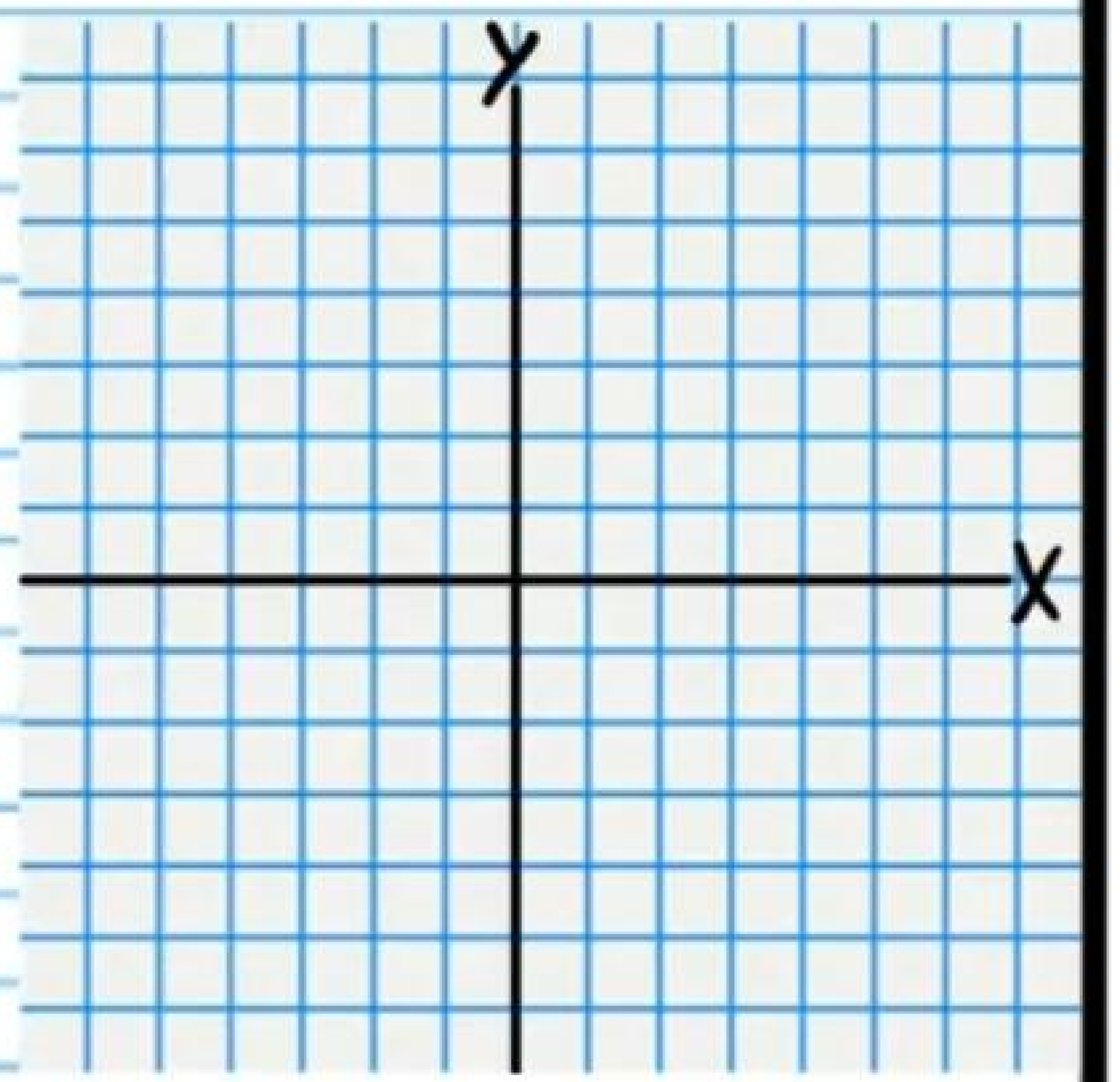
④ $x-3=3(y-2)^2$



(5) $y-4=-(x+2)^2$

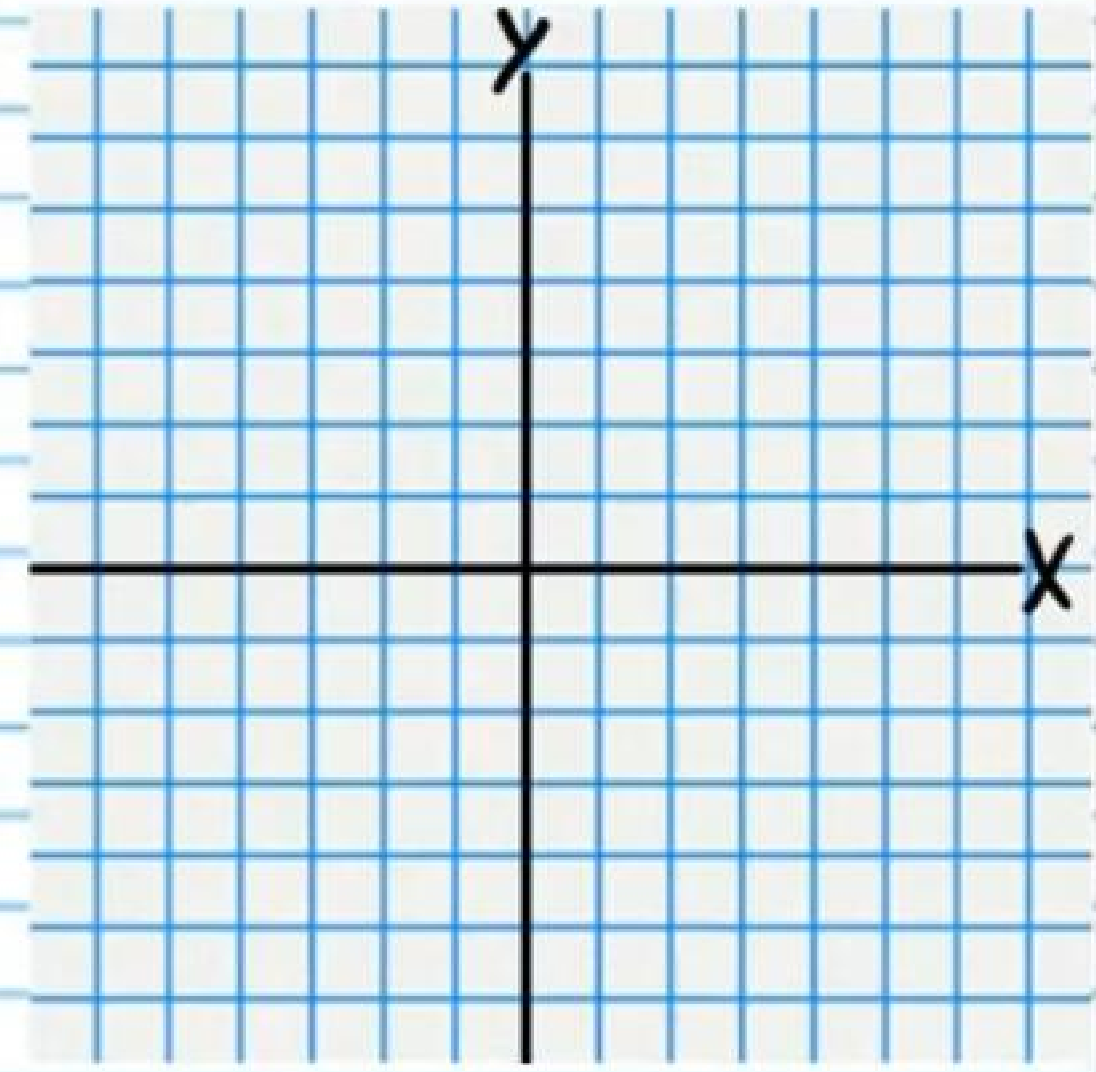


(6) $-\frac{1}{3}y^2=x$

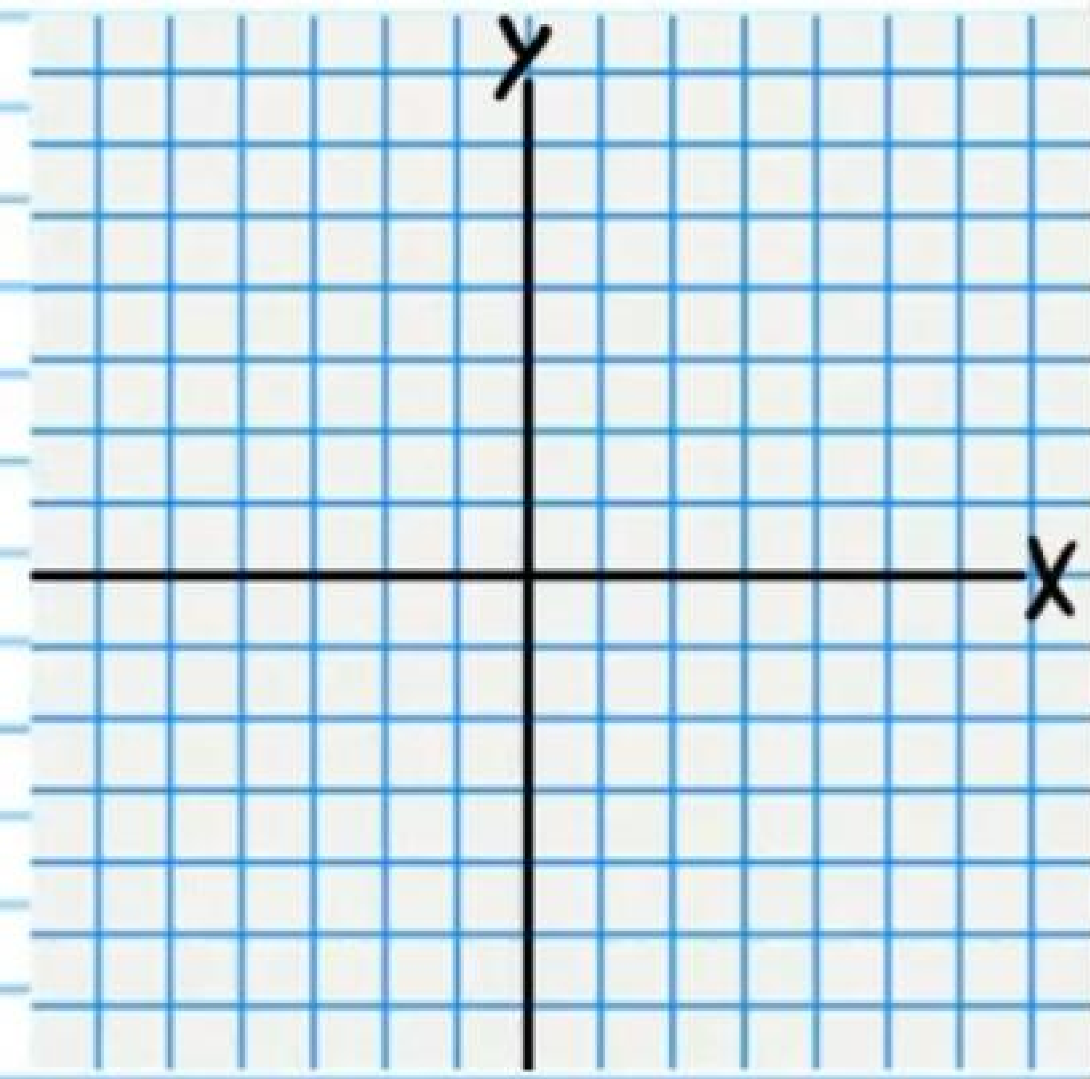


Graph the following equations. Write the coordinates of the foci.

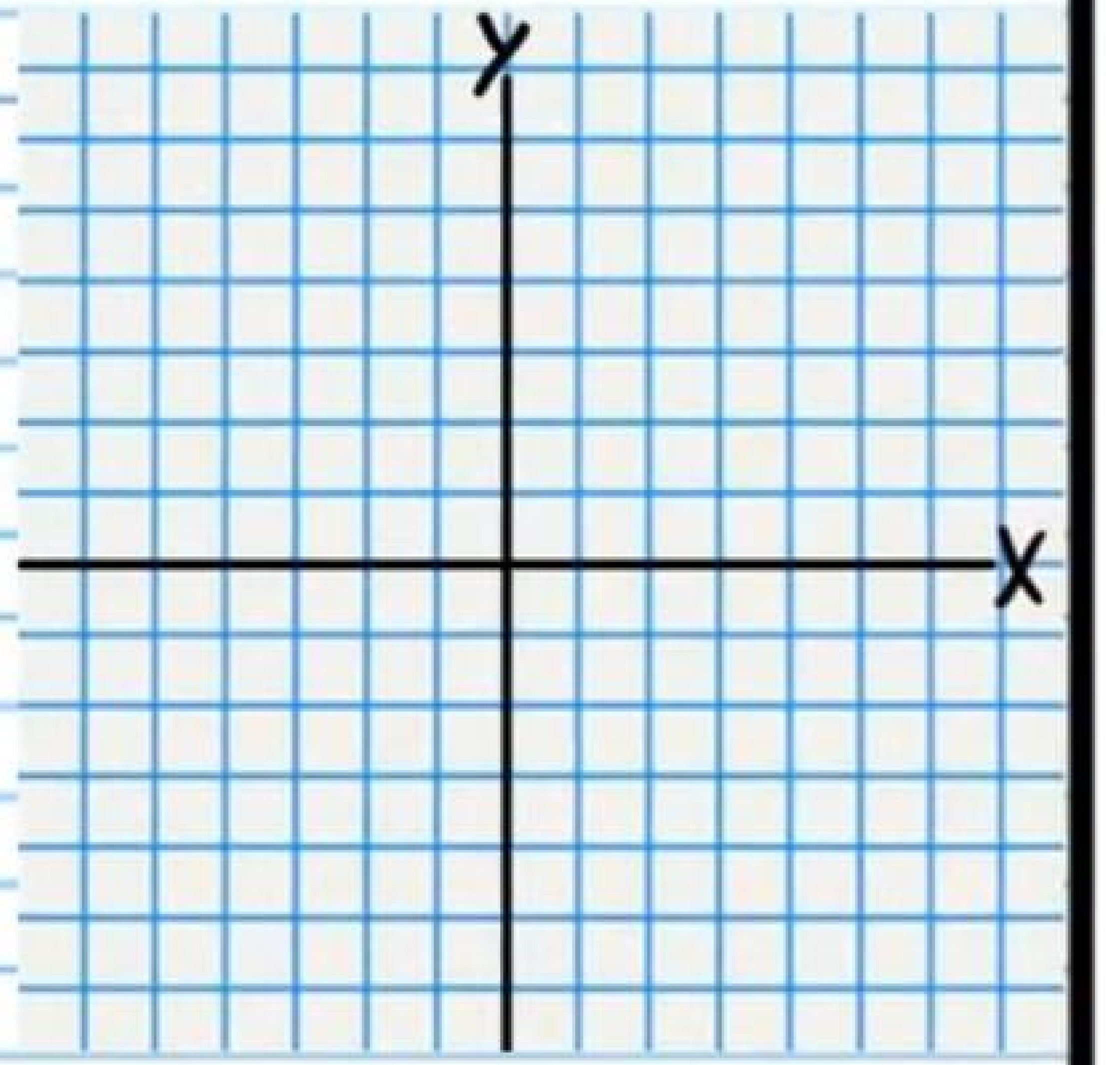
(7) $\frac{(x-4)^2}{4}+\frac{(y-3)^2}{9}=1$



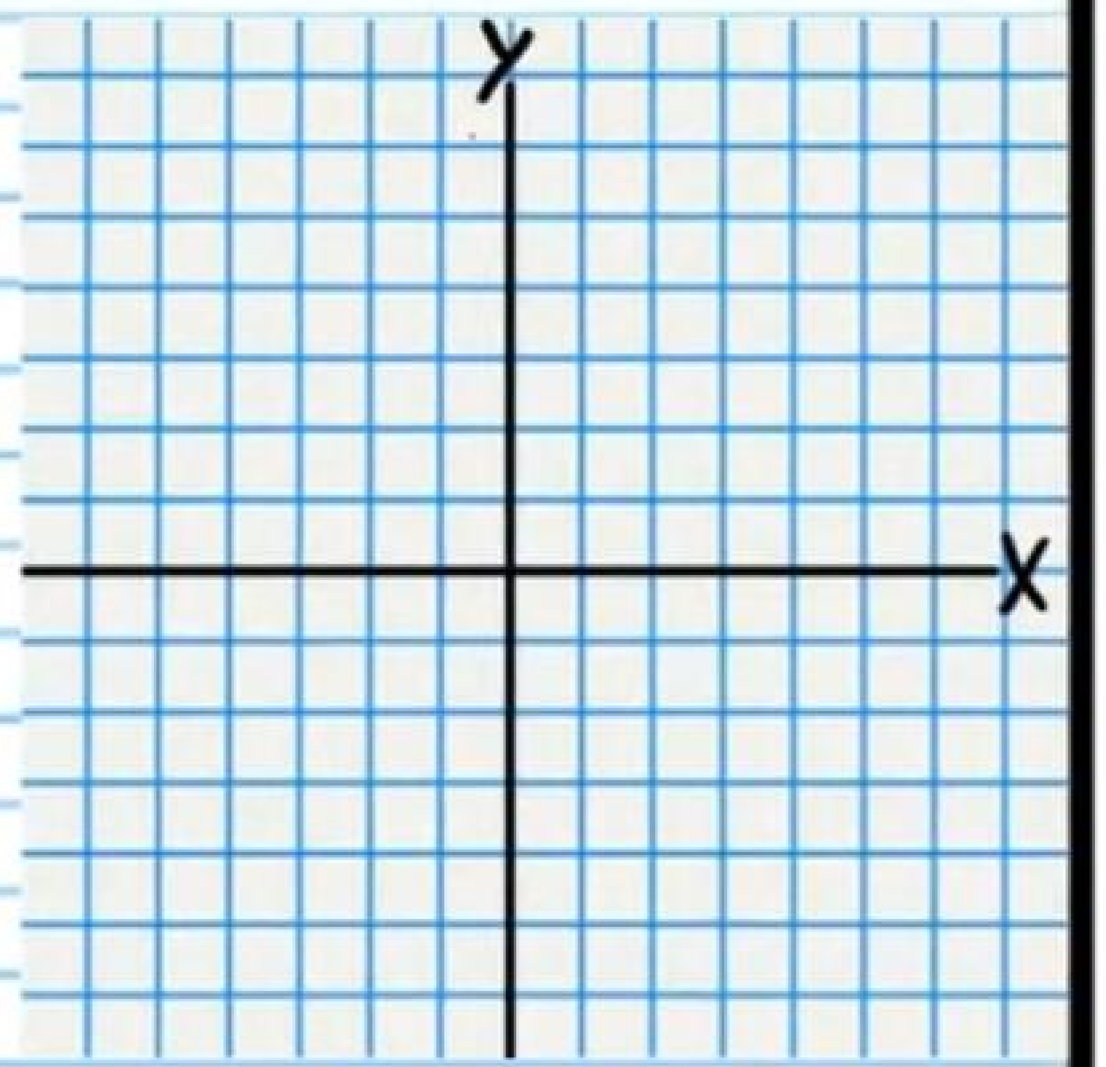
(8) $1=\frac{(x-1)^2}{25}+(y+4)^2$



⑨ $\frac{(x+3)^2}{9} + \frac{(y-1)^2}{16} = 1$

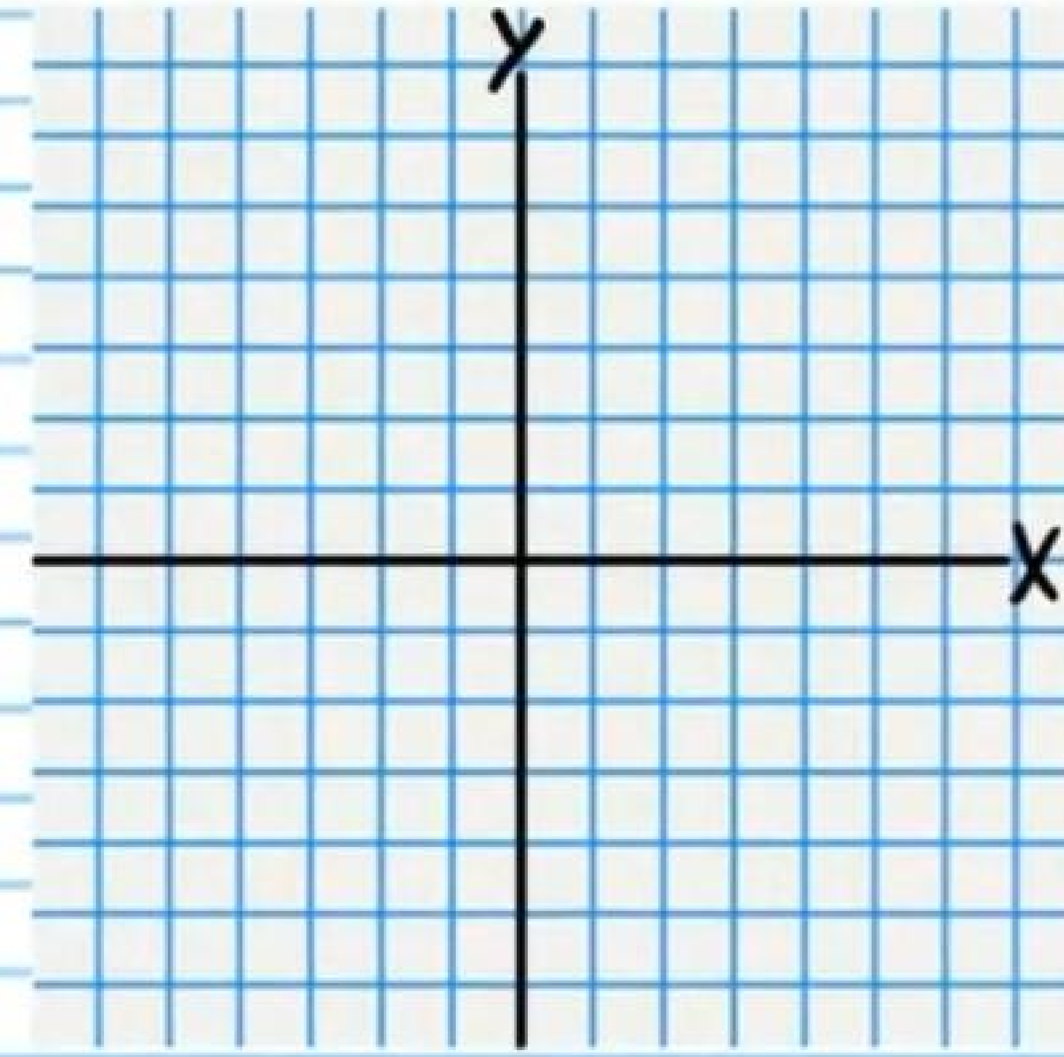


⑩ $1 = \frac{(x+1)^2}{36} + \frac{y^2}{12}$



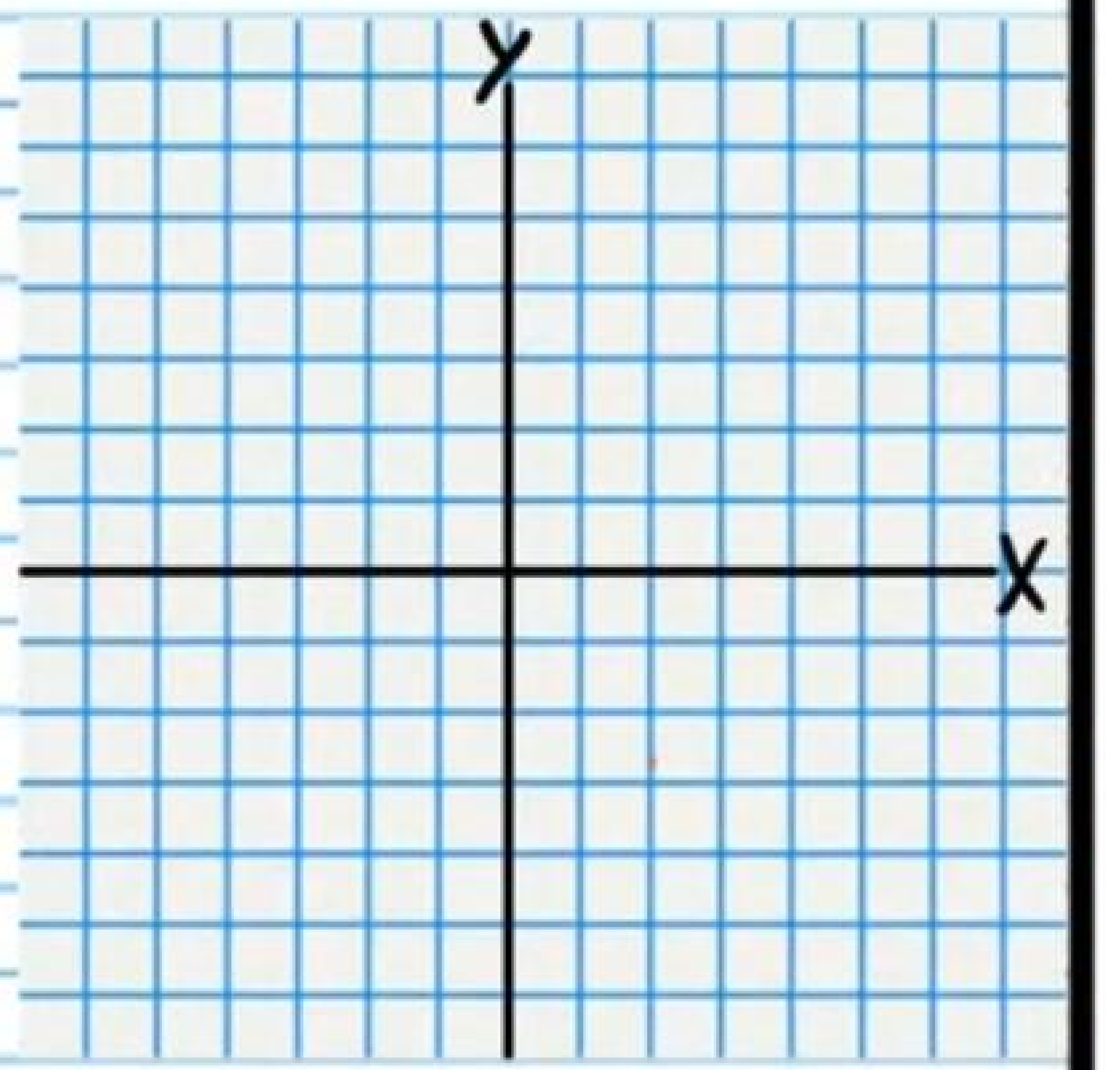
Graph the following equations. Write the coordinates of the foci, draw the asymptotes as dotted lines, and write their equations.

⑪ $\frac{(x+1)^2}{16} - (y+1)^2 = 1$



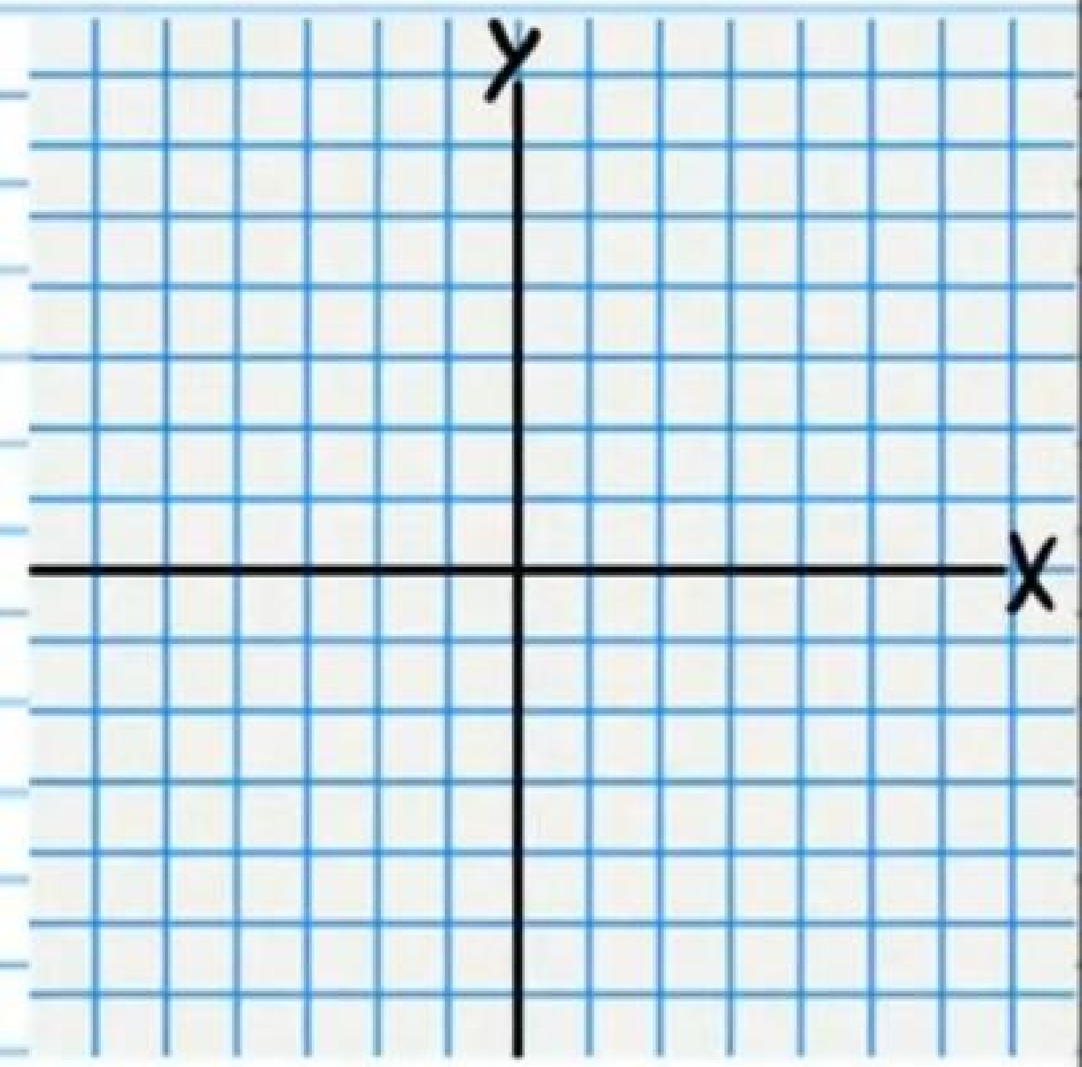
Asymptotes

⑫ $1 = \frac{(y+2)^2}{9} - \frac{(x-2)^2}{9}$



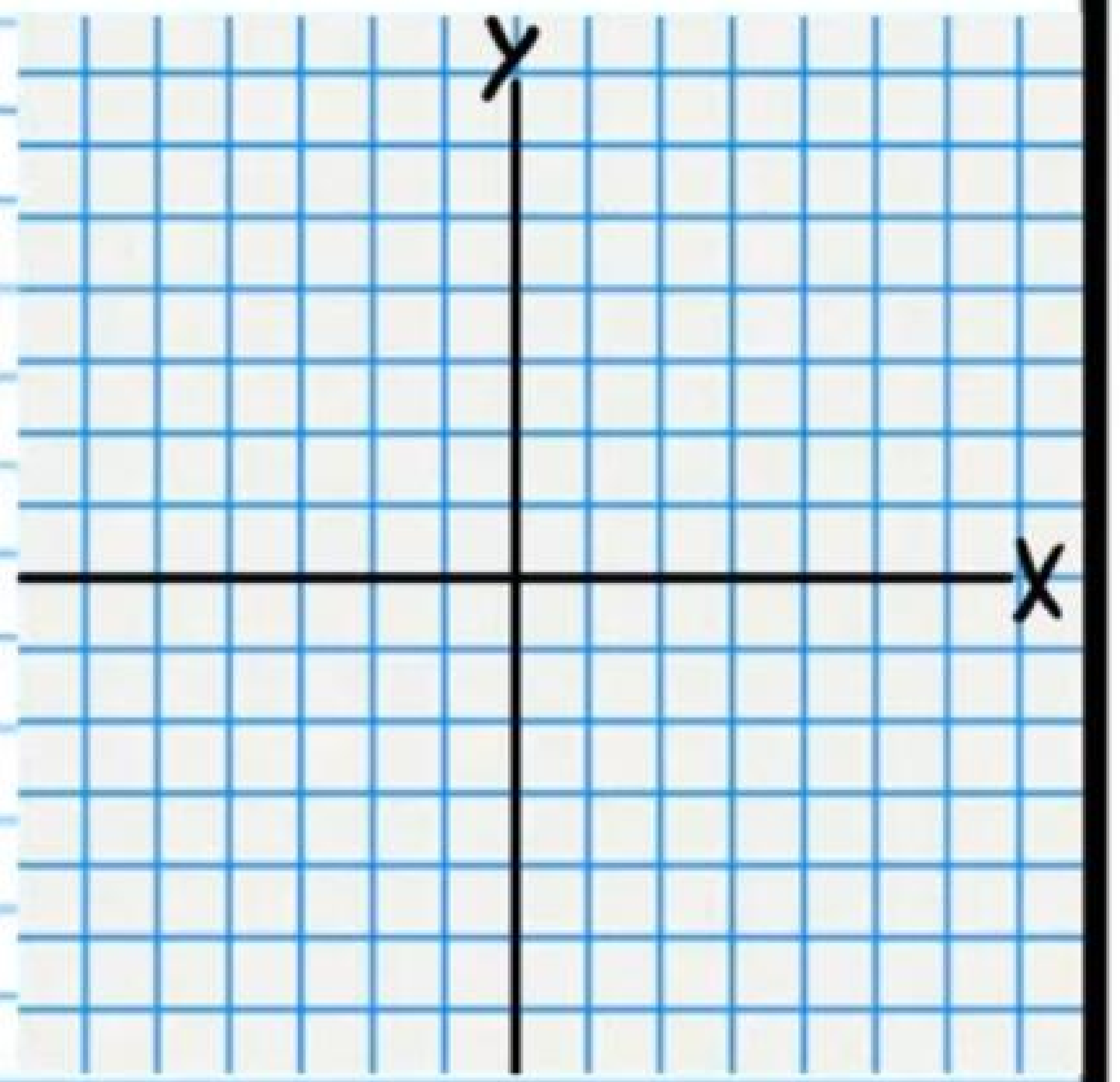
Asymptotes

⑬ $\frac{(x-1)^2}{25} - \frac{(y-3)^2}{8} = 1$



Asymptotes

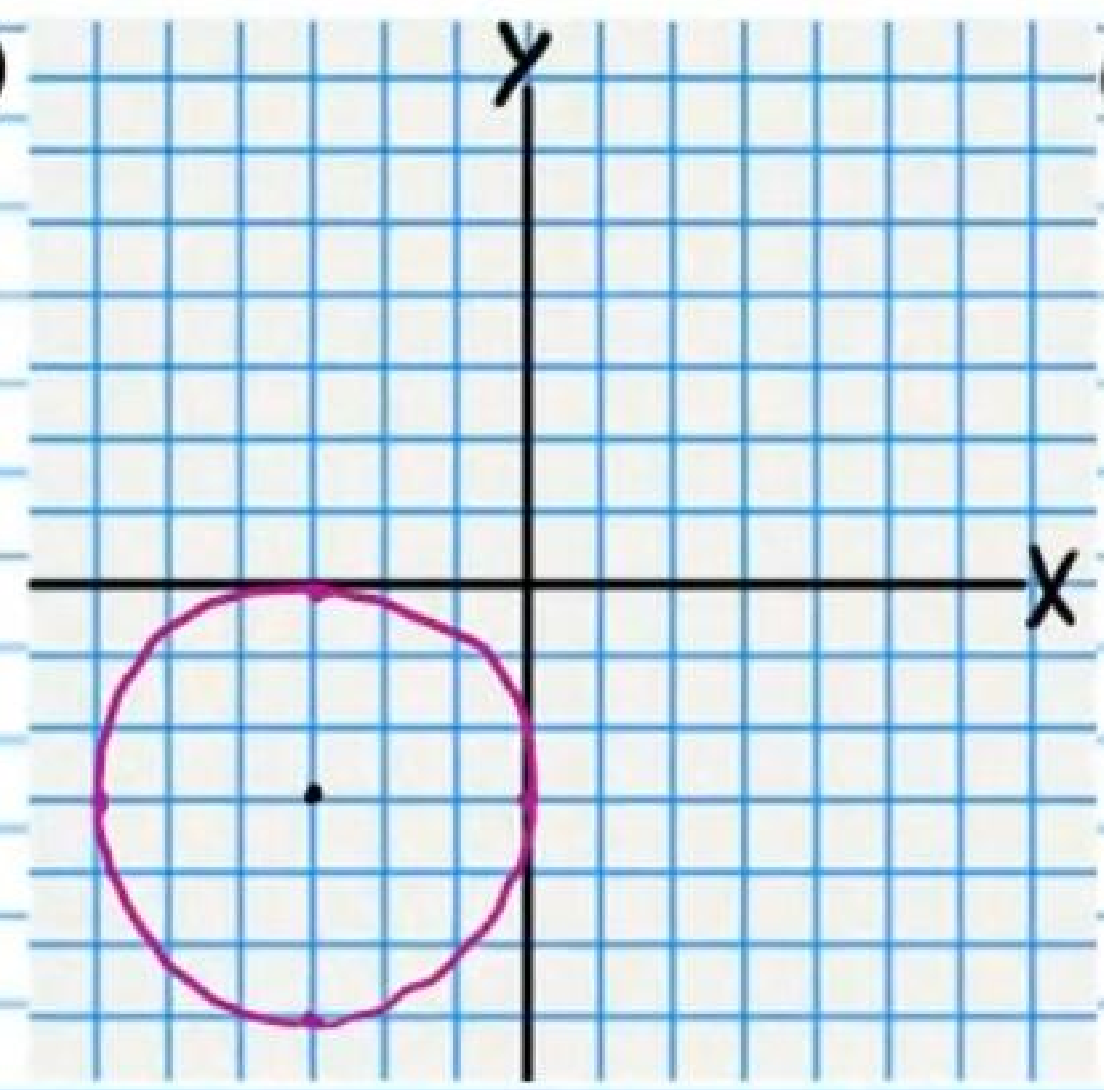
(14) $1 = y^2 - x^2$



Asymptotes

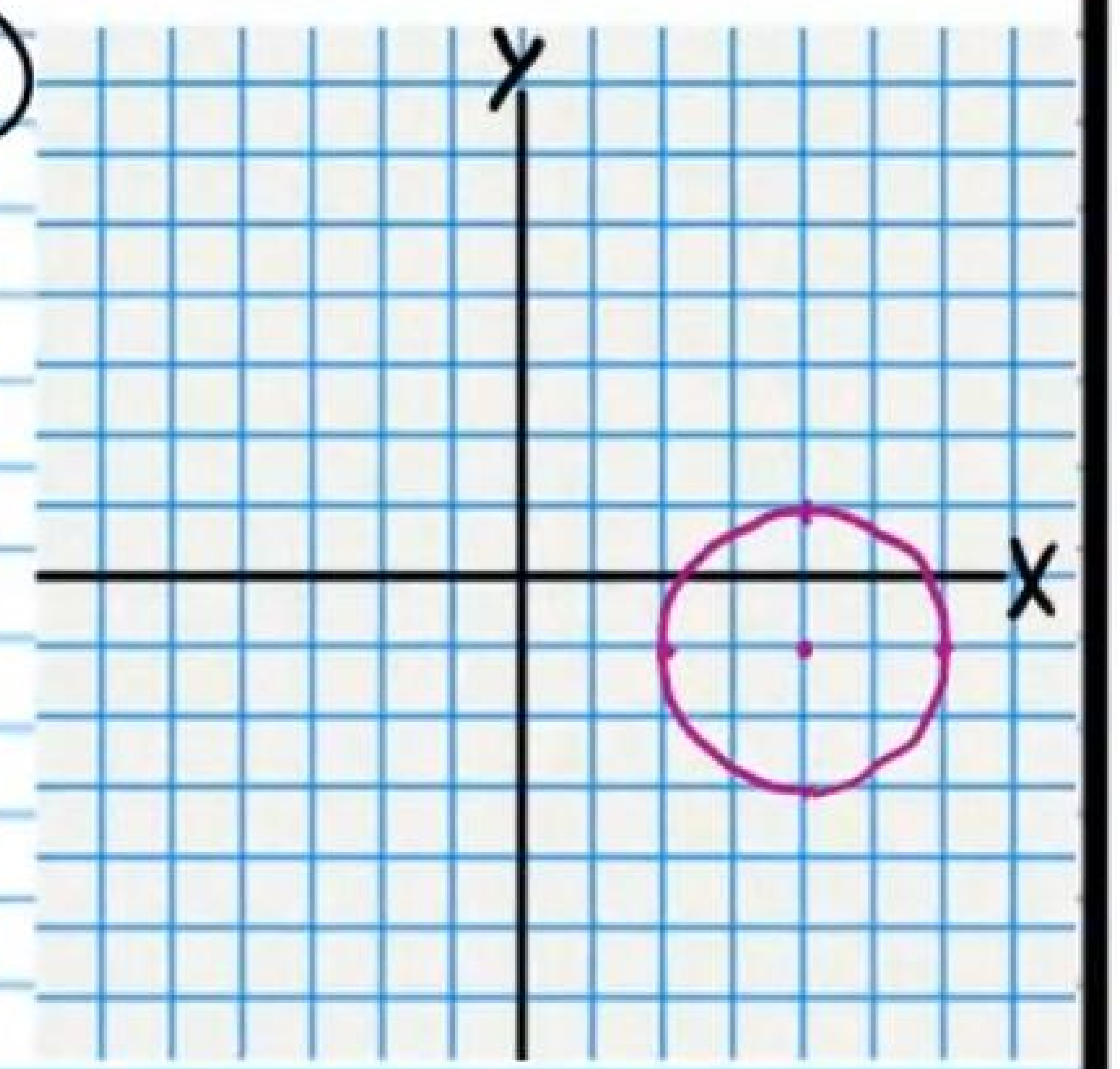
Answers

①



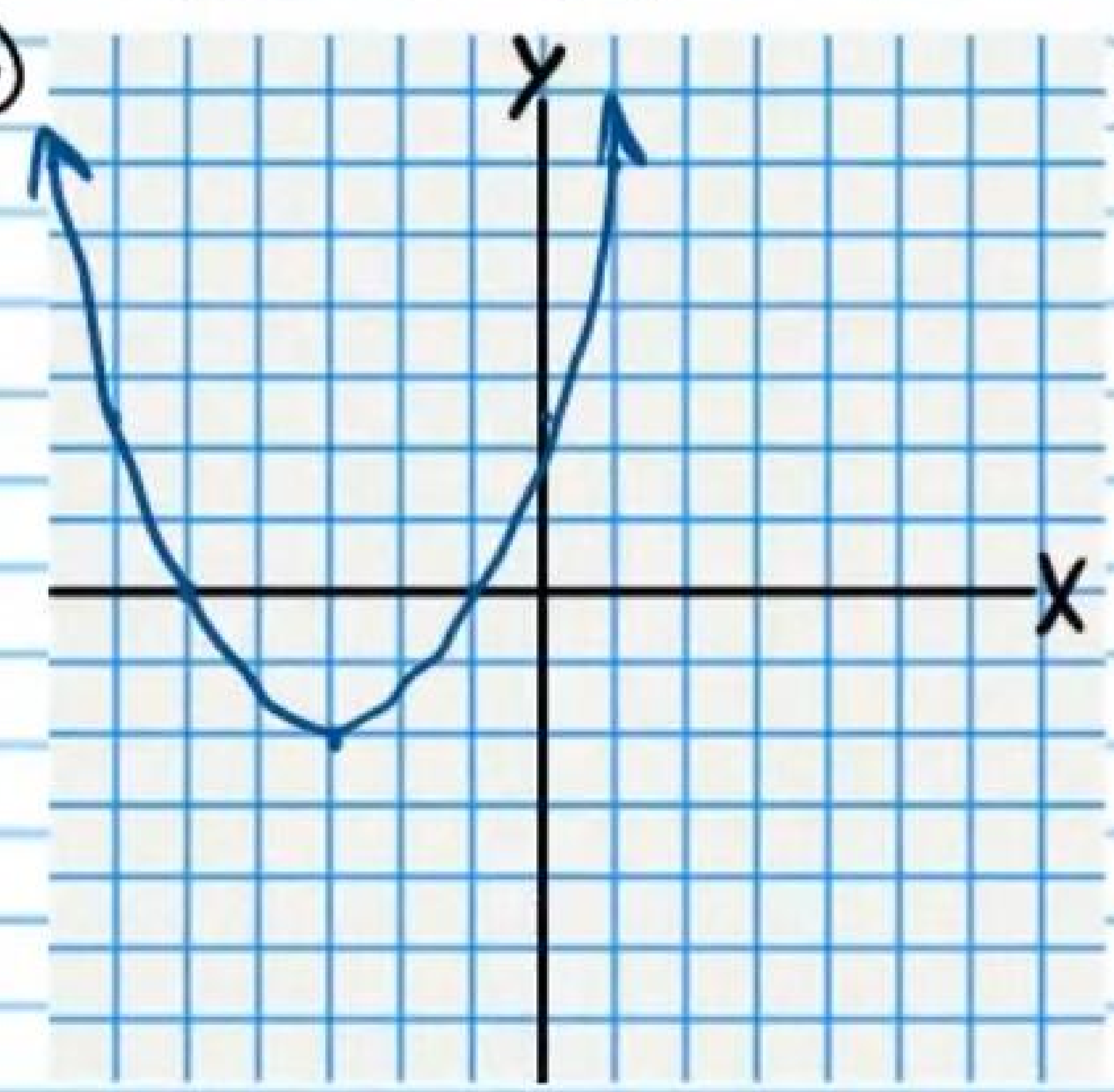
$C(-3,-3)$ $r=3$

②



$C(4,-1)$ $r=2$

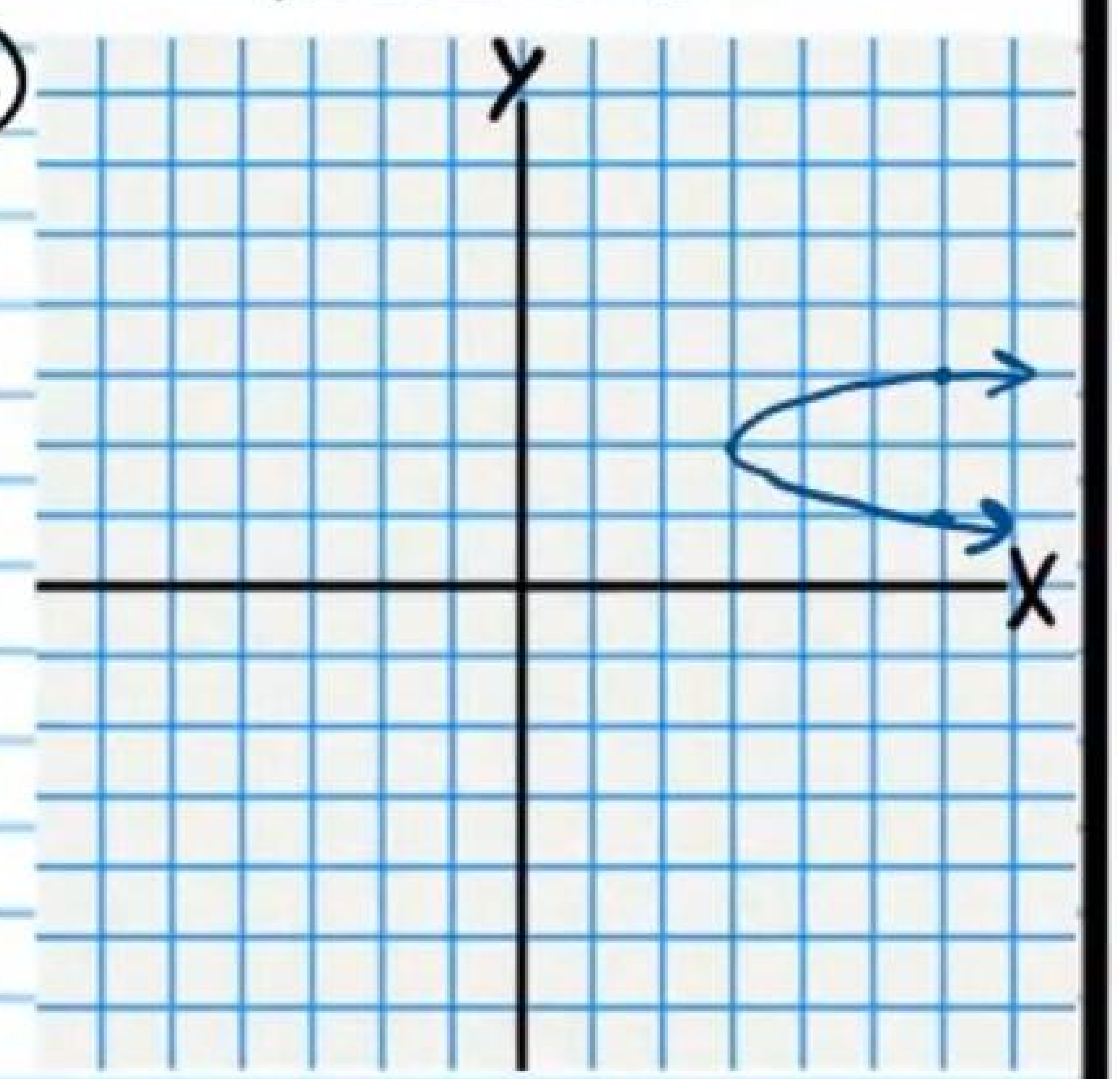
③



$F(-3,-1\frac{1}{2})$

$y=-2\frac{1}{2}$

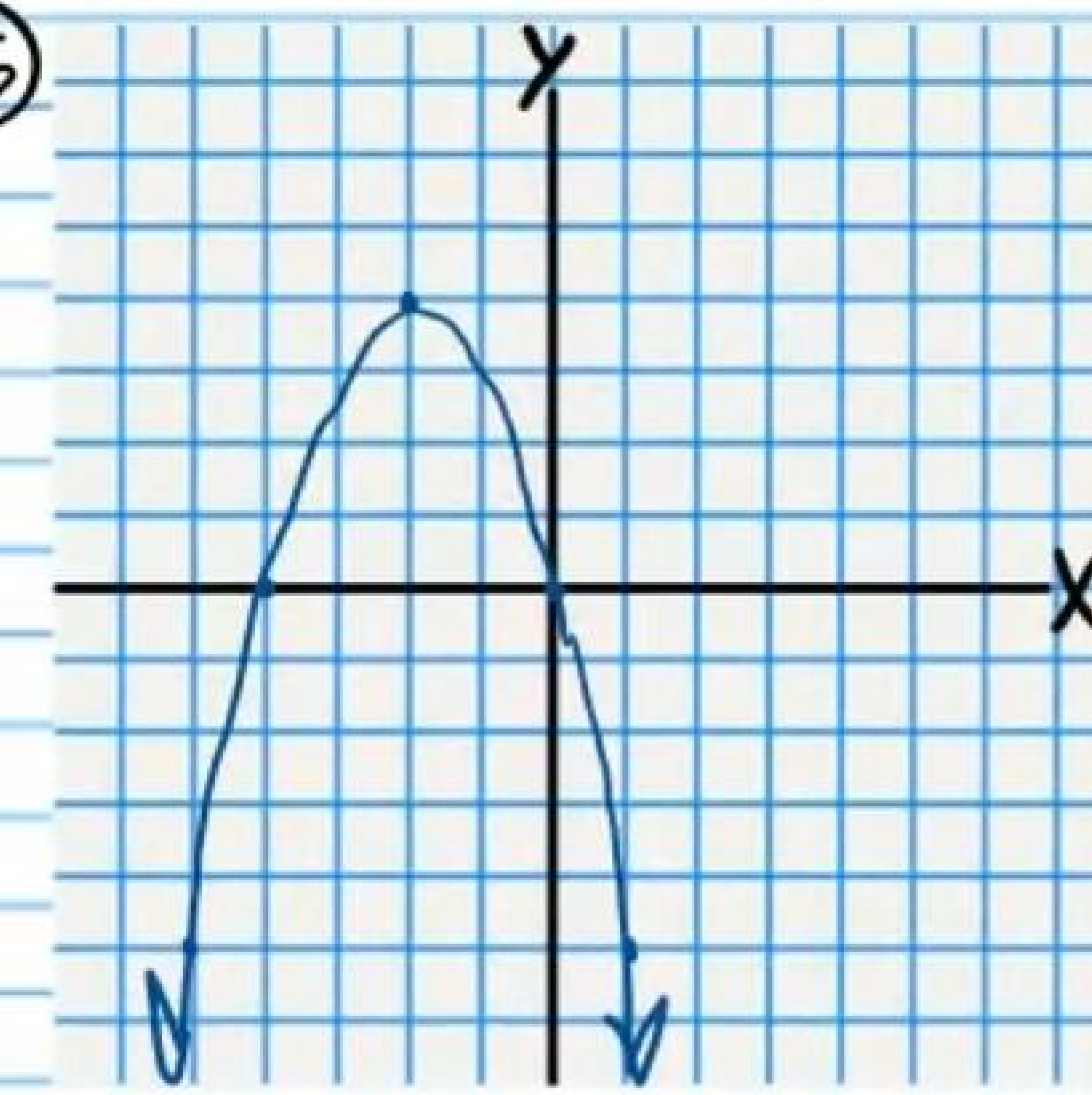
④



$F(3\frac{1}{12},2)$

$x=2\frac{11}{12}$

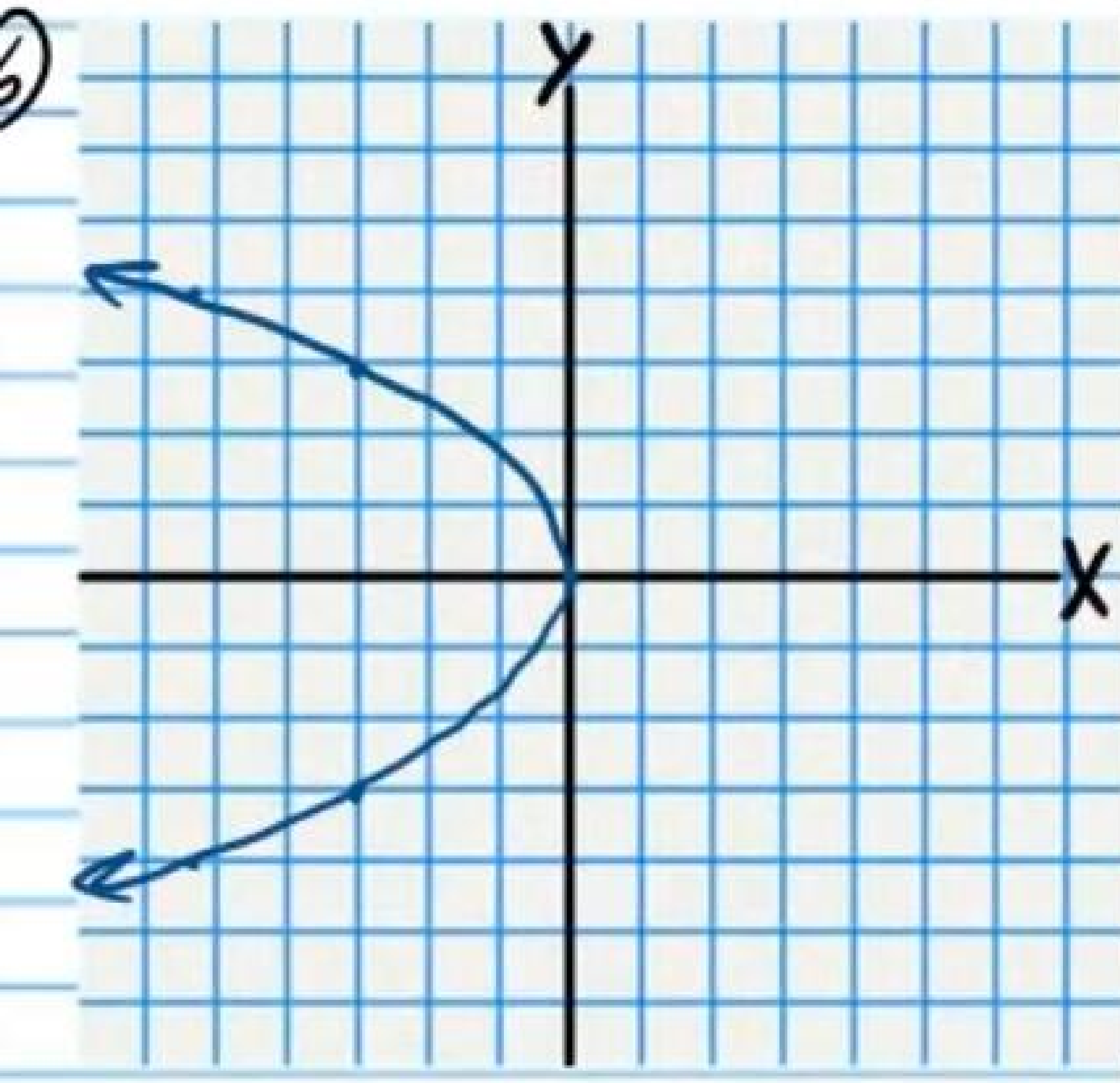
⑤



$F(-2,3\frac{3}{4})$

$y=4\frac{1}{4}$

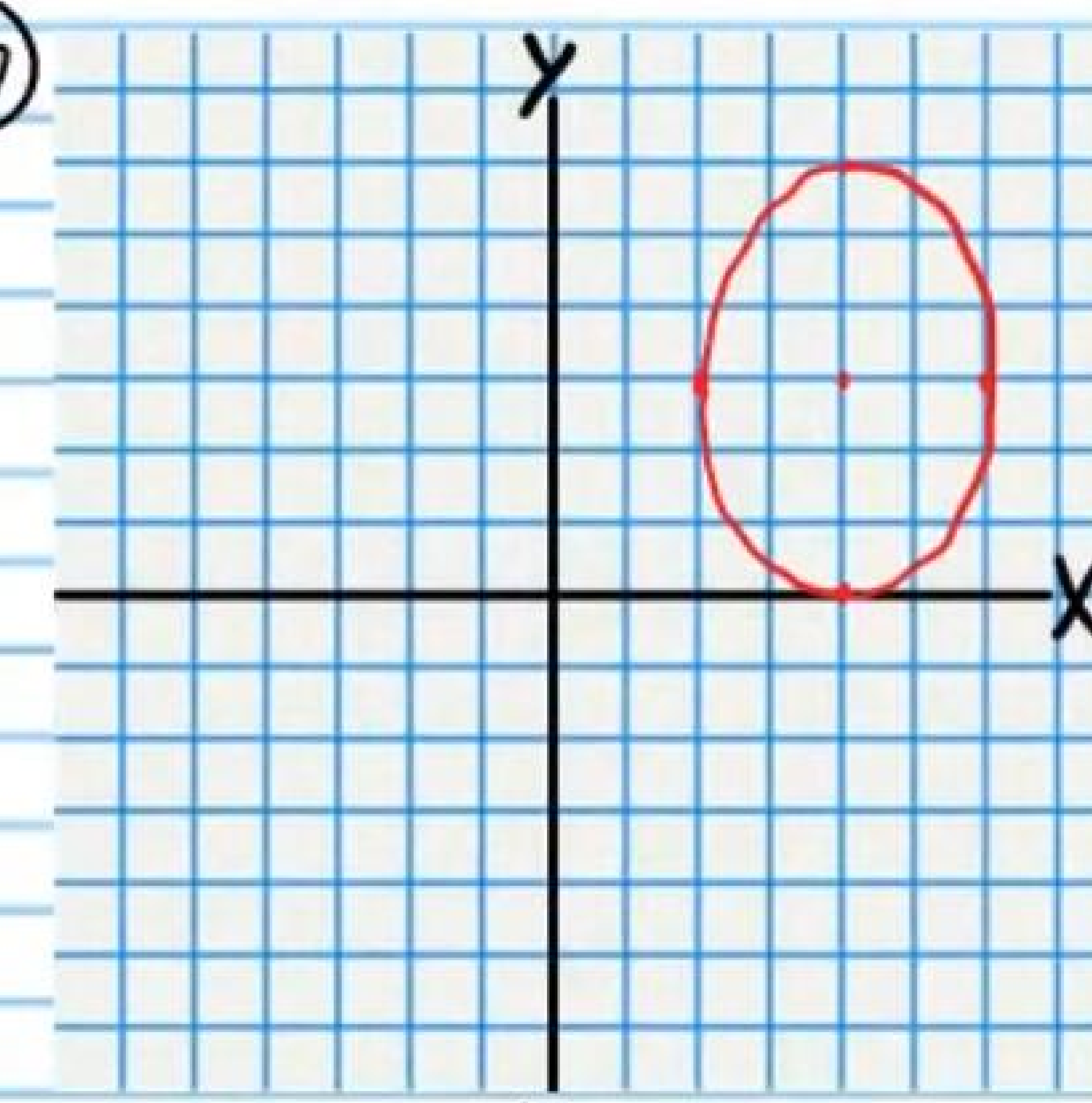
⑥



$F(-\frac{3}{4},0)$

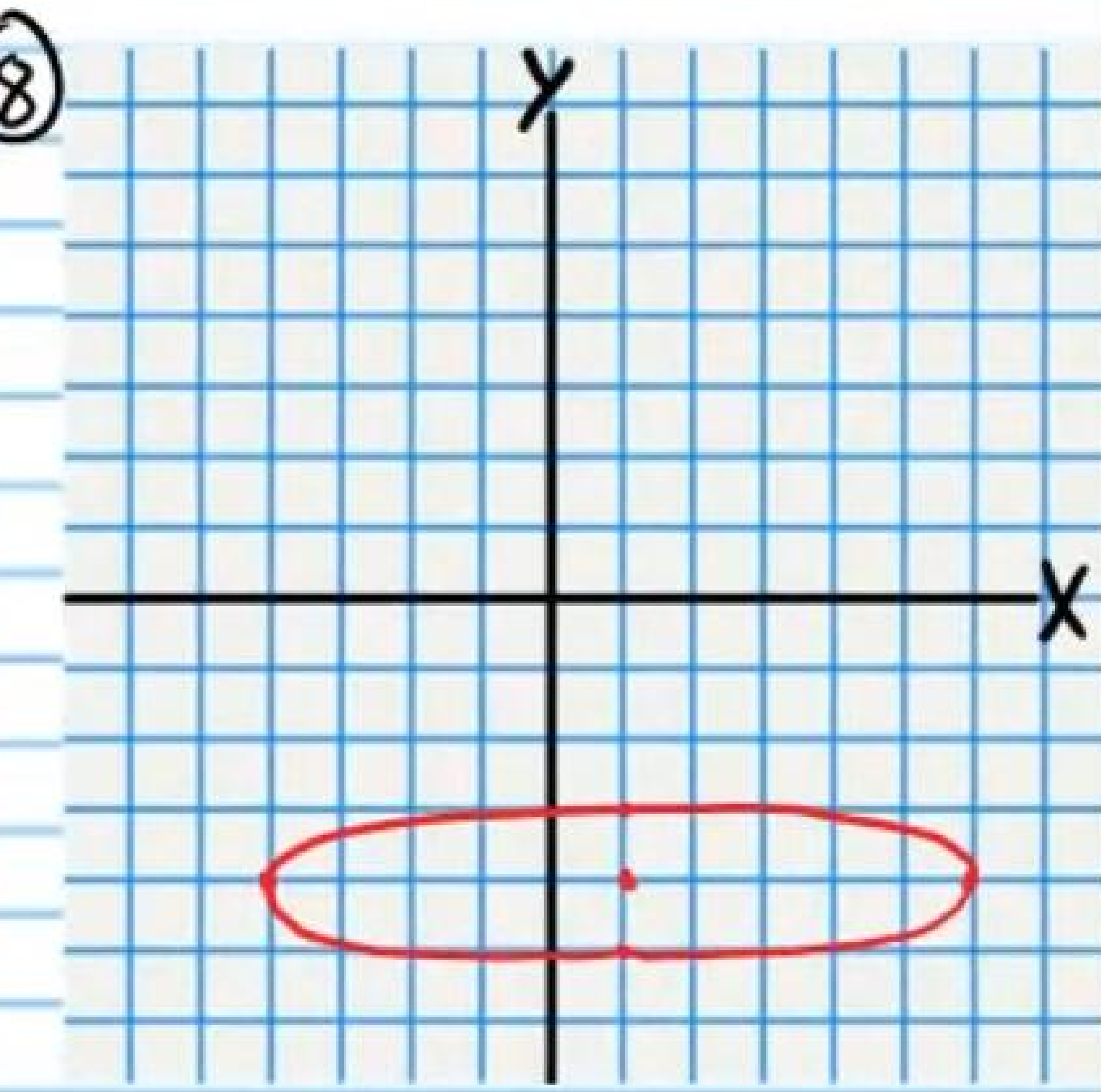
$x=\frac{3}{4}$

⑦



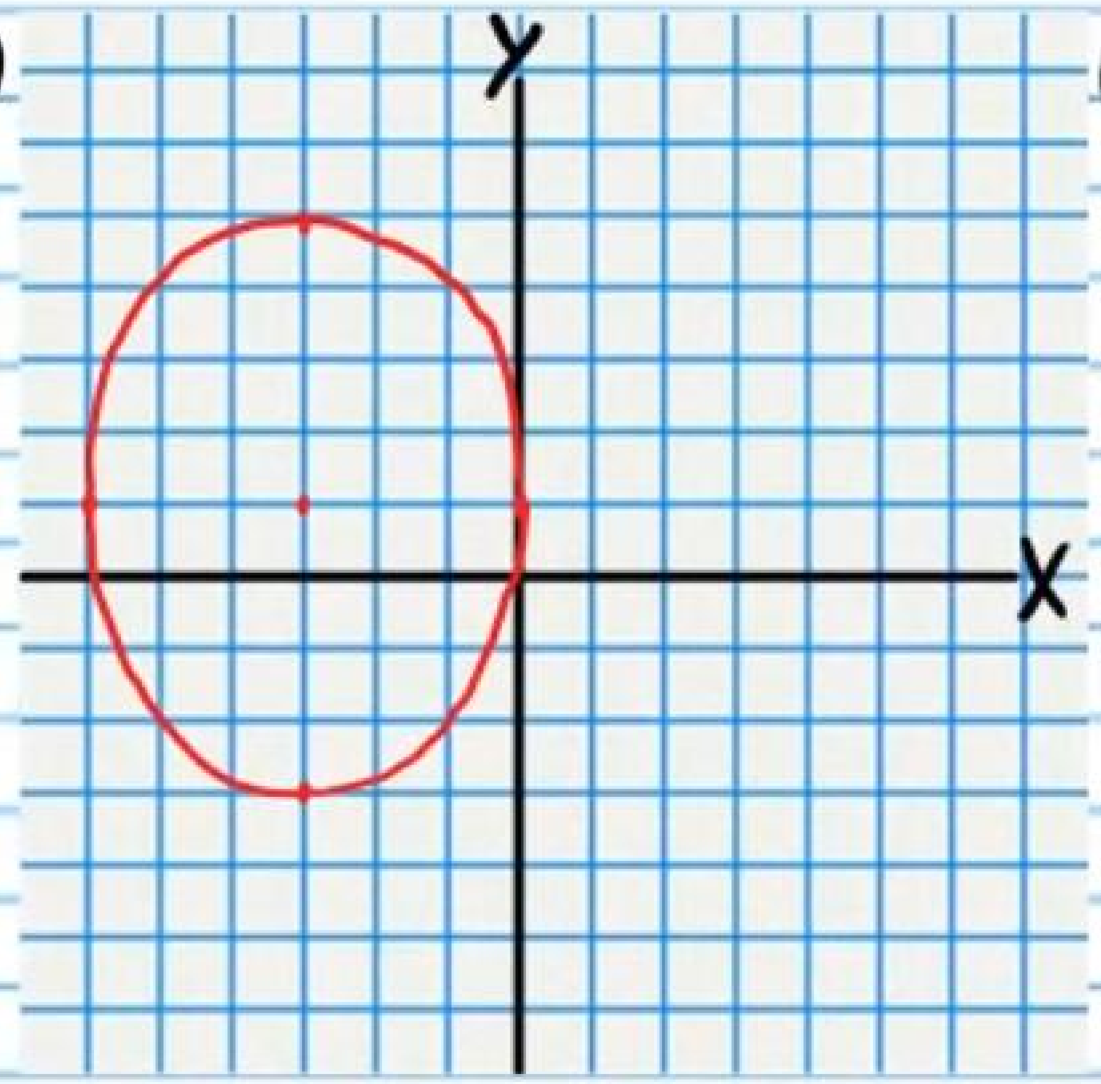
$F_1(4,3+\sqrt{5})$ $F_2(4,3-\sqrt{5})$

⑧



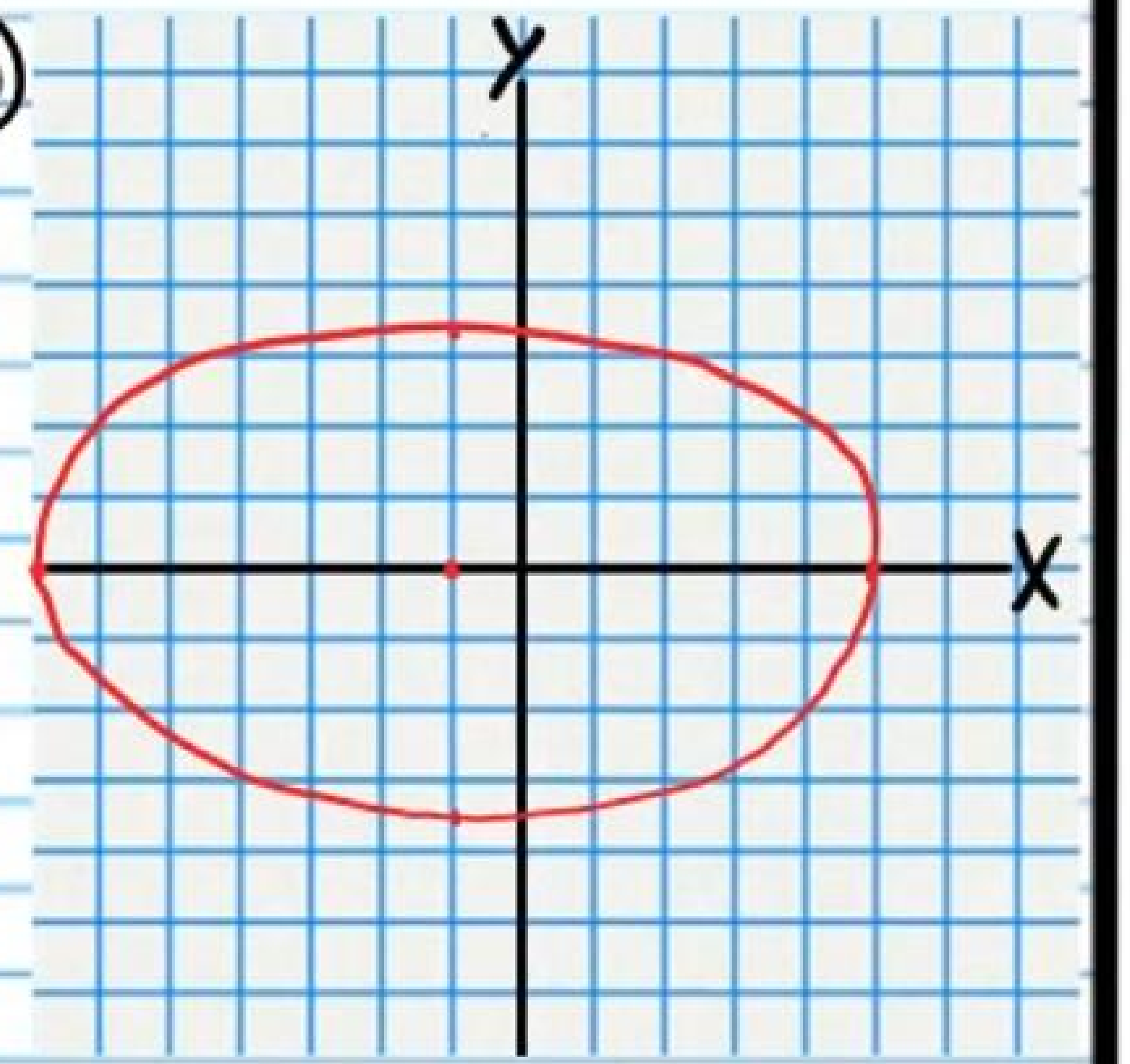
$F_1(1-2\sqrt{6},-4)$ $F_2(1+2\sqrt{6},-4)$

(9)



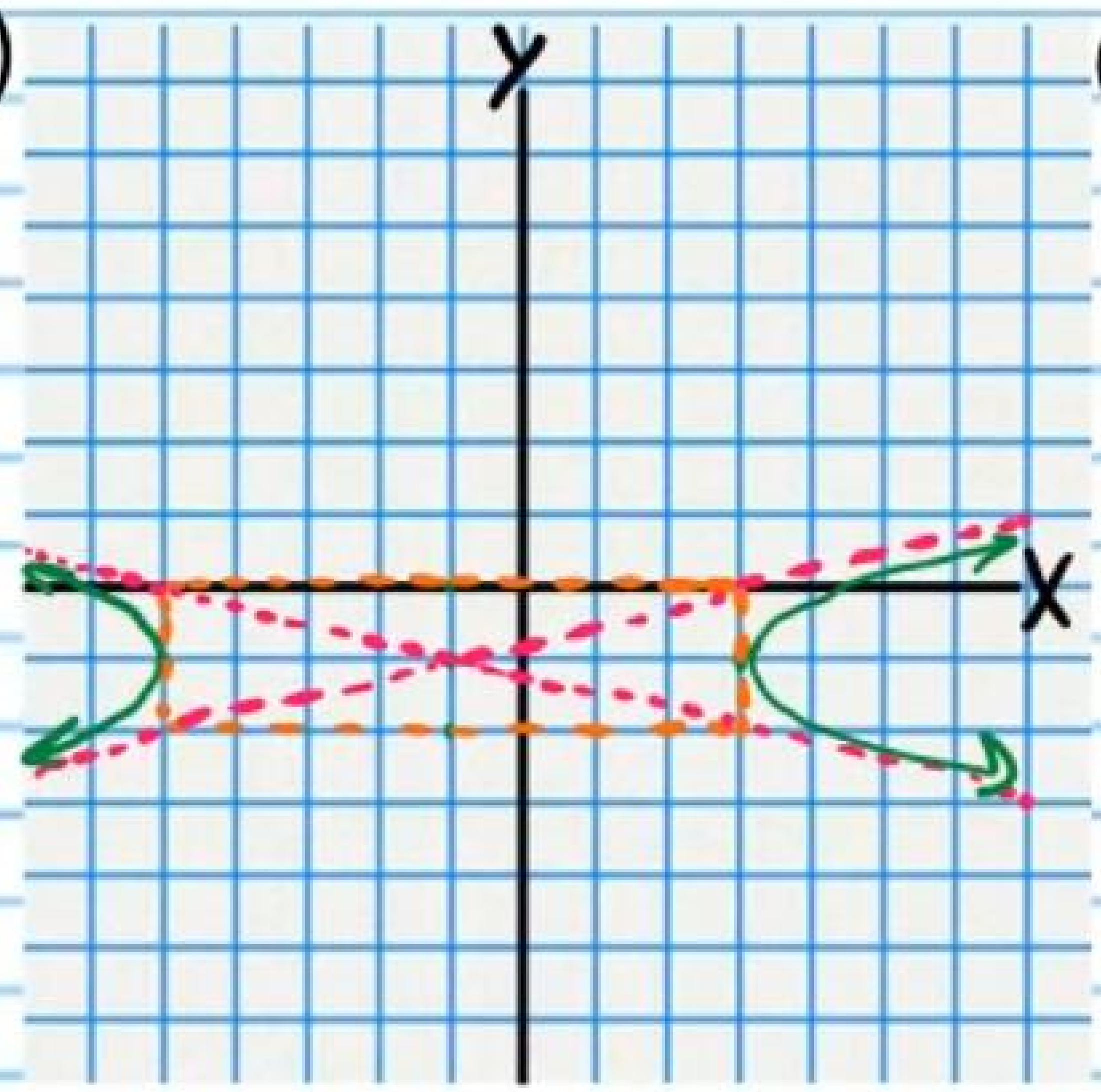
$F_1(-3, 1+\sqrt{7})$ $F_2(-3, 1-\sqrt{7})$

(10)



$F_1(-1+2\sqrt{6}, 0)$ $F_2(-1-2\sqrt{6}, 0)$

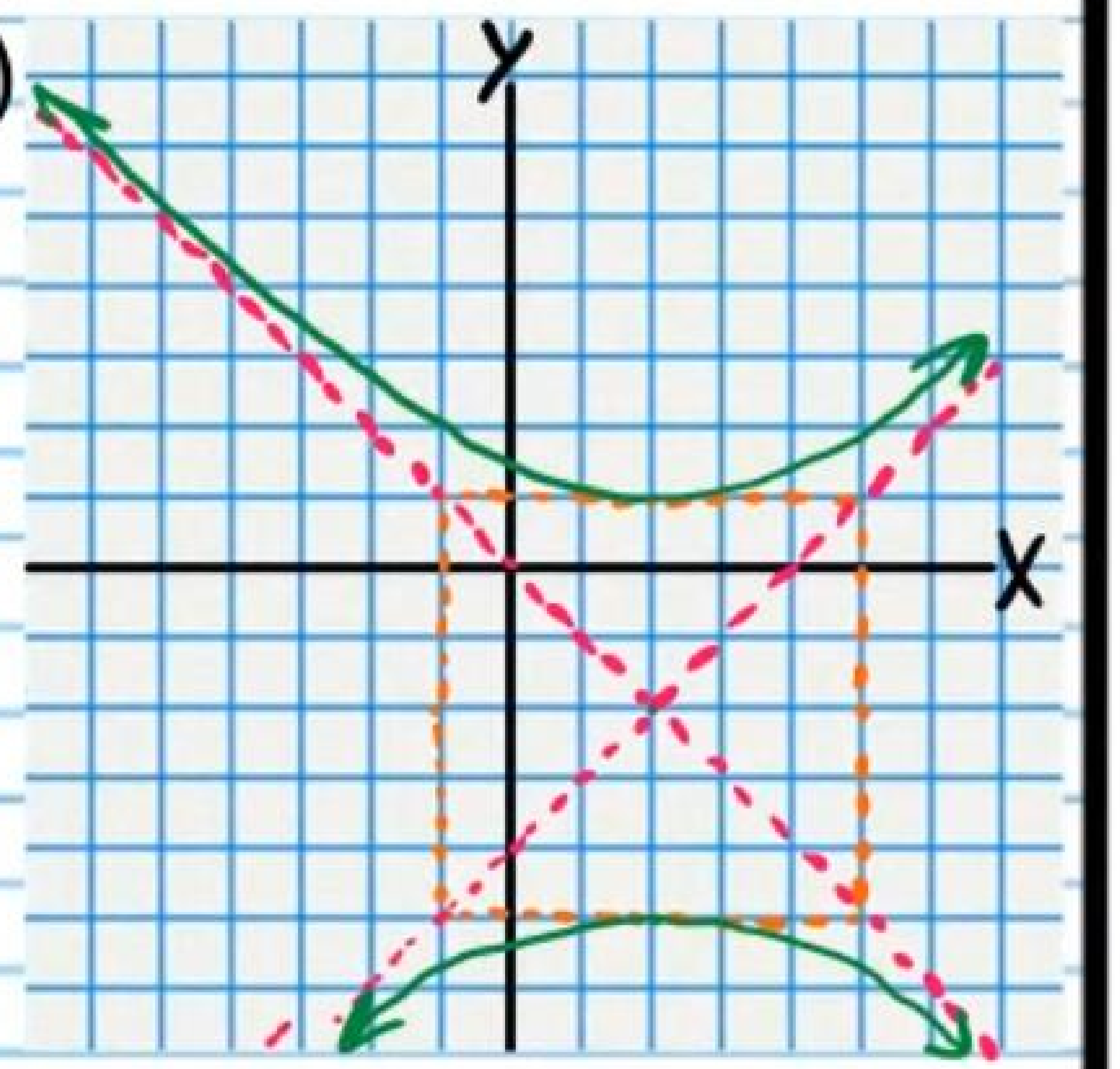
(11)



$F_1(-1-\sqrt{17}, -1)$ $F_2(-1+\sqrt{17}, -1)$

$y+1 = \frac{1}{4}(x+1)$ $y+1 = -\frac{1}{4}(x+1)$

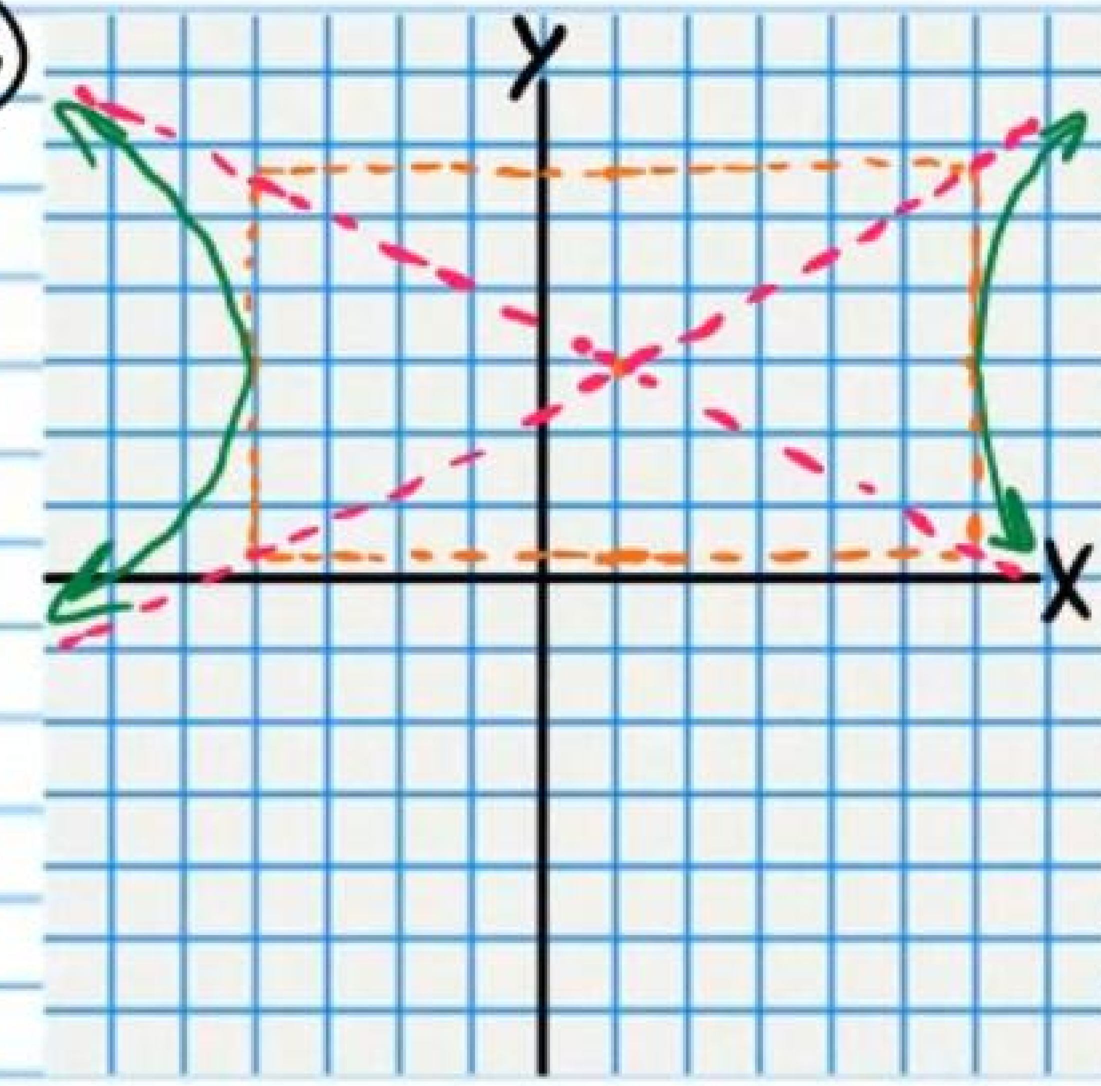
(12)



$F_1(2, -2+3\sqrt{2})$ $F_2(2, -2-3\sqrt{2})$

$y+2 = x-2$ $y+2 = -(x-2)$

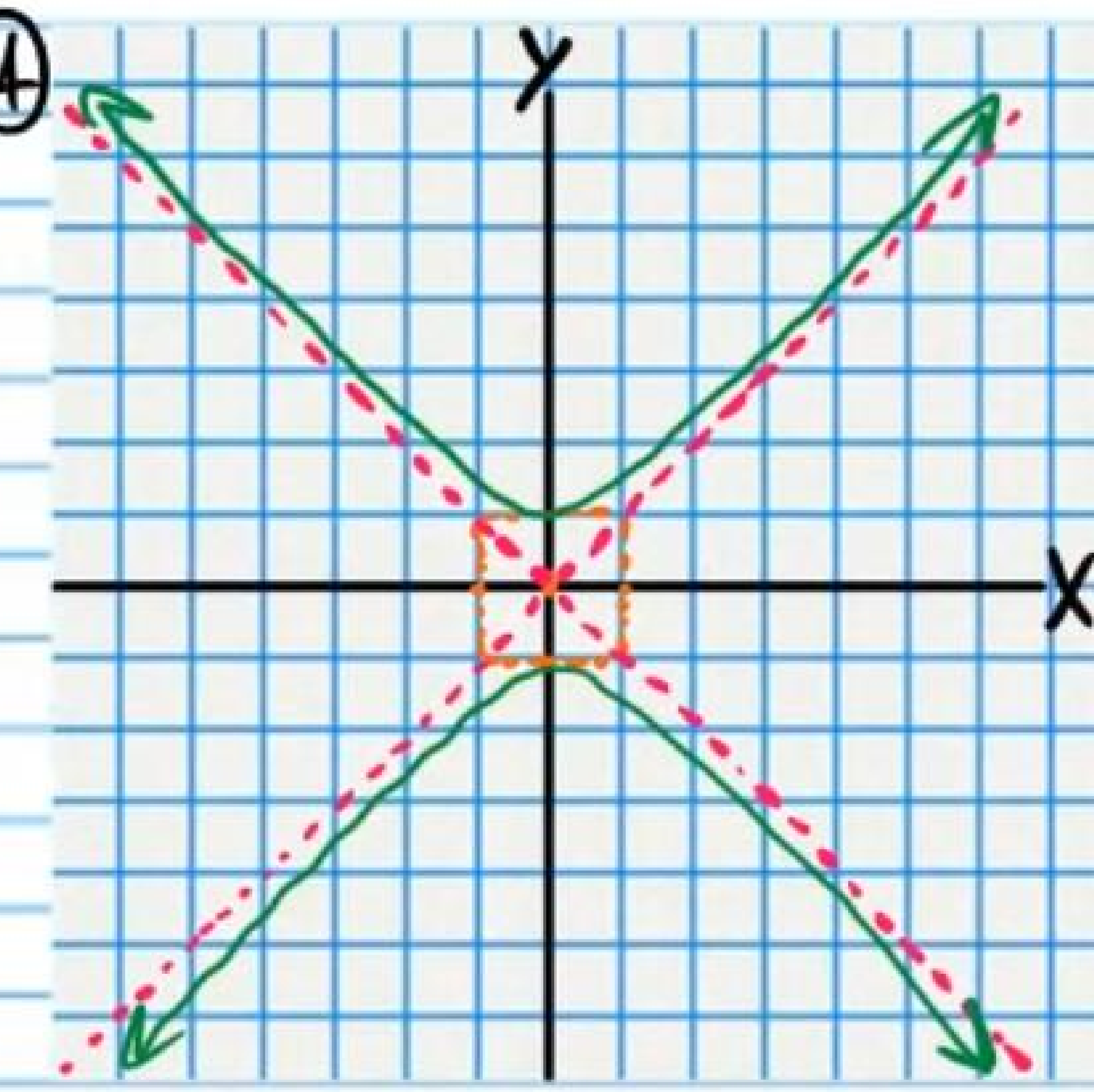
(13)



$F_1(1+\sqrt{33}, 3)$ $F_2(1-\sqrt{33}, 3)$

$y-3 = \frac{2\sqrt{2}}{5}(x-1)$ $y-3 = -\frac{2\sqrt{2}}{5}(x-1)$

(14)



$F_1(0, \sqrt{2})$ $F_2(0, -\sqrt{2})$

$y = x$ $y = -x$